

Is “Categorical Imperative” Metaversal?: A Kantian Ethical Framework for Social Virtual Reality

Eyup Engin Kucuk

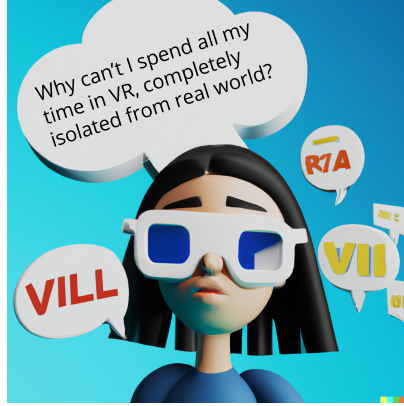
Khoury College of Computer Sciences
Northeastern University
Boston, MA, USA
e.kucuk@northeastern.edu

Caglar Yildirim

Khoury College of Computer Sciences
Northeastern University
Boston, MA, USA
c.yildirim@northeastern.edu



(a) Persona A



(b) Persona B



(c) Persona C

Figure 1: Personas for case studies. The images for the personas are AI-generated and were produced by feeding the descriptions of the personas into DALL-E 2 (<https://openai.com/dall-e-2/>).

ABSTRACT

The increasing adoption of social virtual reality (VR) environments for socializing and collaborating with others has led to a growing concern about ethical issues in these immersive environments. Beyond the introduction of some practical guidelines, theoretical work on this topic has been scant. In this paper, we propose an ethical framework for social VR based on Kant’s Theory of Morality. In so doing, we argue that the Kantian concept of *categorical imperative* does apply to social VR, that the reality of VR is not different from the reality of real life, and that what is morally unacceptable in real life *is* and *should be* unacceptable in social VR. In our framework, we provide three principles that can aid users and developers of social VR environments in reasoning about ethical issues in social VR, while advocating for more theoretical approaches to addressing the issue of VR ethics in human-computer interaction circles.

CCS CONCEPTS

• **Human-centered computing** → HCI theory, concepts and models.

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA ’23, April 23–28, 2023, Hamburg, Germany

© 2023 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-9422-2/23/04.

<https://doi.org/10.1145/3544549.3585911>

KEYWORDS

Kantian ethics, social VR, VR ethics

ACM Reference Format:

Eyup Engin Kucuk and Caglar Yildirim. 2023. Is “Categorical Imperative” Metaversal?: A Kantian Ethical Framework for Social Virtual Reality. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA ’23)*, April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3544549.3585911>

1 INTRODUCTION

Virtual reality (VR) technologies present an alternative environment where humans can carry on most of their daily real-world activities. In VR, it is possible to socialize with friends and family, attend meetings, share social environments with others, shop, play a game with others, and so on. When we consider the rapid developments in VR, it seems that it is going to present a more useful, affordable, and comfortable medium that captures most of our needs in life. Beyond that, as Chalmers [7] suggests, the following might be the case: “In centuries past, families often faced a decision: ‘Should we emigrate to a new country to start a new life?’ In centuries to come, we may face an equivalent decision: ‘Should we move our lives to a virtual world?’ As with emigration, the reasonable answer may often be yes” (p. 15). These facts bring forth several discussions about VR, but two of them are philosophically interrelated: “What is the nature of “reality” in VR?” and “Do we need ethical guidelines in VR?”. It is not easy to answer one of them

without answering, or at least considering, the other, because our attitude, as researchers and users, about ethical considerations are determined by our perception of the nature of reality. To illustrate, if one thinks that “VR is a kind of game world where we can do whatever we want”, then it may be easy to conclude that there is no need for any form of ethical framework or guideline. On the other hand, if one assumes that “VR is a form of reality that is identical, or at least similar, to our real-world reality”, then there must be some considerations about the limits of our actions in VR. The need for ethical reasoning is more dire in social VR environments that enable users to interact with others through embodied avatars [20], because these environments offer immersive and realistic interactions that can have lasting effects on both parties [25].

Several attempts have been made to develop ethical guidelines, determine the risks and dangers, and present a code of conduct for both researchers and users [4, 6, 10, 12, 19, 27, 29]. However, the current literature is limited by two main factors. First, they do not consider the above-mentioned questions in relation to one another. Second, they present some practical precautions for minimizing the risks (which is extremely important), but they do not propose an ethical framework or provide guidance for ethical reasoning in social VR. For the former, it is obvious that there is no clear-cut answer to the reality question, and we can even say that we do not have a determined answer for the nature of our real-world reality. However, examining the perception of reality in VR (e.g., device-gap problem, the difficulty of simulating tactile sensation and so on) is not enough for developing ethical reasoning. The lack of or the future possible success in these problems only changes our perception of virtual reality and does not make much difference in understanding this reality. For the latter, in connection with the first, we might develop some form of code of conduct or practical guidelines by examining the existing technologies, but they are almost certainly subject to change. What is needed is a method of ethical reasoning that could serve as the theoretical foundation for our practical principles and guidelines for social VR. Therefore, there is no such necessity to answer the reality question first and then formulate ethical regulations. We can present some answers to both questions by considering the nature of experience in VR and its relation to our experience in the real world.

This is exactly the purpose of the present paper: We provide a framework for assessing the ethical nature of our experience in social VR environments, in which multiple users have the opportunity to interact and communicate with one another online through 3D avatars. In our opinion, Kantian ethics is a proper framework to serve as a theoretical base for our ethical reasoning in social VR, for in social VR environments, users engage in a shared virtual experience with and through which they interact in a manner similar to physical environments through their virtual representations. We do not claim in any way that it is the only rationally plausible ethical framework. Rather, our objective is to encourage the HCI community to think about VR ethics from a theoretical lens rooted in one of the most influential schools of thoughts in philosophy and to offer guidelines for current and potential ethical issues in VR. The contributions of this paper are theoretical in nature and include (i) a theoretical framework based on Kant’s Theory of Morality with three principles and (ii) a method of ethical reasoning for decision making in social VR, illustrated with three example case studies.

2 RELATED WORK

2.1 Normative Ethics

While our framework is based on Kant’s theory of morality, it is important to note that other ethical theories, especially normative theories, can be applied to social VR. Therefore, in this section, we briefly describe their basic tenets to facilitate our discussion of why Kant’s theory is directly applicable to social VR environments.

Normative ethics is the branch of philosophy concerned with morality and how individuals should approach and behave in moral situations. Theories of normative ethics provide principles for formulating what is right and wrong and rules of thumb for assessing the extent to which one’s actions or decisions are moral [23, 24]. Prominent normative ethical theories include consequentialism, virtue ethics, and deontology (duty ethics). Kant’s theory of morality falls under deontology and is elaborated in the rest of the paper.

In consequentialism, which is also referred to as teleological ethics, the morality of an action is determined by its consequences or outcomes, and an action is deemed good if it yields good consequences [9]. Consequentialism emphasizes the goal-directedness of actions and approaches their morality from this perspective by evaluating the moral worth of the outcomes of an action instead of honing in on the means used to achieve those outcomes [26]. Hence, consequentialism is rather subjective, as outcomes that are deemed good for an individual may not be deemed as such for others.

In virtue ethics, the emphasis is on the moral character of individuals instead of consequences [9]. Virtue ethics argues that ethical behavior is attainable through the cultivation of virtuous character traits, such as compassion, honesty, and fairness [9]. Once individuals uphold and practice these traits, they can engage in moral behaviors, as these virtues guide them in avoiding immoral behaviors [11]. Given its focus on virtues or character traits and its subjective nature, virtue ethics provides no direct actionable guidelines for approaching moral situations, limiting its applicability to complex issues.

In deontology, or duty ethics, the morality of an action is determined by the extent to which it accords with universal obligations or rules of moral agents [9]. Specifically, at the center of Kantian duty ethics is the duty to honor universal human rights and to fulfill generalizable moral obligations. Because of the emphasis on the universalizability of duties, Kant’s theory of morality addresses the subjectivity limitation of consequentialism and virtue ethics. Rather than relying on the subjective goodness of outcomes or virtues that may change from one person to another, Kant emphasizes the need to approach moral situations based on universal principles [16]. If the action cannot be generalized so that it is good, beneficial, or right for everyone, then it is not considered moral [15]. We, therefore, argue that the simplicity and consistency of Kant’s approach make it more appropriate and applicable for developing a common ethical understanding that could be generalized to the vast variety of issues in social VR. While consequentialism and virtue ethics offer valuable perspectives, their focus on outcomes and individual character can yield more complex and variable moral judgments, making it difficult to establish consistent ethical guidelines for social VR.

2.2 VR Ethics

In one of the leading papers related to VR ethics, Slater et al. [27] base their analysis and arguments on the golden rule of reciprocity, namely “that which is hateful to you, do not do to your fellow”. Slater et al. also argue that acting immorally or disrespectfully to other people in VR environments might cause psychological harm to people and we should avoid that. In addition, they focus on more practical ethical issues associated with immersive environments, namely motion sickness, information overload, intensification of experience, and re-entry to the real-world. They examine these issues in relation to the notion of *superrealism*. In other words, the risks and problems are considered in light of the reality of experience and its possible dangers.

In another paper, Madary and Metzinger [19] follow a similar path with more concrete suggestions. The attention of the paper is mostly on the nature of the experience in VR environments and they are clear that this article is on the applied ethics of VR. They present a concrete list of recommendations for both researchers and users. In this sense, this paper presents a useful code of conduct as a way to minimize the mentioned risks. Moreover, Spence’s [28] attempts to formulate a meta-ethical framework for virtual worlds based on Alan Gewirth’s “Principle of Generic Consistency”. By extensively explaining and reconstructing Gewirth’s theory, Spence [28] tries to offer an ethical framework especially for reconciliation of conflict of rights and freedom in virtual worlds.

There are of course several papers focusing on the ethics of VR, but these three papers represent the three main approaches in considering the ethics of VR. First approach is shared by Slater et al. [27] and Madary and Metzinger [19]. They focus on the nature of the experience in terms of its reality and assess the ethical issues from this point of view. The second approach is to present all potential or actual problems, some of which are not specific to VR environments and are valid for all forms of technology, such as data privacy. This approach is useful in that it enables us to see the big picture of ethical issues in VR. The last approach is represented by Spence’s paper [28]. It presents an extensive ethical framework and relates it to the virtual worlds. Although all these approaches certainly have merits for exploring the subject matter, they also carry some shortcomings. The first shortcoming is that directing the attention to the reality question based on empirical data. As indicated earlier, the reality question is essential, but grounding ethical reasoning on empirical data might not be sufficient. The reason is that even in the real world, where we are sure immoral acts could cause personal and public harm, we still need to refer to an ethical framework so as to develop a response to the question “why do I need to act morally?”. Most of the risks and problems in VR environments are not specific to virtual experiences and are in fact valid for all sorts of technologies. There is no doubt that contemplating these risks and problems is necessary and useful, but it is not sufficient for developing an ethical framework. Madary and Metzinger [19] clearly state that “scientists must understand that following a code of ethics is not the same as being ethical. A domain-specific ethics code, however consistent, developed, and fine-grained future versions of it may be, can never function as a substitute for ethical reasoning itself.” (p. 19). For ethical reasoning, we need to think beyond rules, regulations and codes of conduct.

The second shortcoming is related to focusing mostly on the doctrine of rights rather than the doctrine of virtues and in the case of VR ethics, this distinction [15] is mostly overlooked. There are some ethical risks and concerns, such as data privacy and identity theft, which must be considered under the class of duties of right. We believe that directing our attention to issues that are specific to VR environments and under the class of duties of virtue is a more effective way for developing an ethical reasoning. In other words, it would be prudent and fruitful to focus on internally commanding duties (the duties of virtue) in formulating the fundamentals of an ethical framework for social VR. There are two main reasons for this claim: First, the issues, which can be classified as the duties of rights, are valid almost for all technological mediums. Putting them into the center of discussion would not bring a new perspective for assessing the ethical nature of VR environments. Second, we mostly agree on the ethical value of these problems, and they require some legal regulations. Theft will always be a part of the VR world, as it is always a part of the real world. Although we all agree that theft is not ethically permissible, it is not possible to eradicate or totally stop these kinds of actions. We are not saying that considering these problems is unnecessary. What we are trying to say is that proposing a form of ethical reasoning requires thinking beyond the practical legal issues and trying to reveal what rationality commands us to do in every circumstance.

3 KANT’S THEORY OF MORALITY

In this section, we provide an overview of Kantian ethics. We begin by acknowledging that it is not easy to summarize Kant’s ideas on morality without referring to his ideas on theoretical philosophy. However, we try to limit this section to the fundamentals of Kant’s moral theory.

In his philosophy, Kant makes a clear distinction between two branches of philosophy: theoretical and practical. While theoretical philosophy focuses on the questions of what is and what can we know, practical philosophy engages with the question of what ought to be. In other words, theoretical philosophy deals with the problem of knowledge and practical philosophy deals with the problem of action. Similar to his approach to theoretical philosophy, on the subject of morality, Kant firstly looks for the main principle through which we can decide what is permissible, obligatory, and prohibited. For Kant, this principle cannot be based on something empirical such as happiness, because happiness is a subjective criterion ([16], 4:395). There can be infinitely different ideas on how to achieve happiness. That’s why, in Kant’s view, there is no way to have a common law that governs the actions of every human being for achieving prudential ends. Morality must, therefore, be based on a law which is normative for all rational agents ([16], 4:389). This kind of law cannot be found in the empirical realm but only in the principles of reason and rationality. By following the rationality of human beings, we can achieve a universal rule by which we can assess all of our actions. Additionally, this approach eliminates the problem of subjectivism because the rational thinking capacity is common for all humans: “Finally there is one imperative that, without being based upon and having as its condition any other purpose to be attained by certain conduct, commands this conduct immediately. This imperative is categorical.” And this single

categorical imperative is to “act only in accordance with that maxim through which you can at the same time will that it become a universal law” ([16], 4:421). What the categorical imperative means is that when an individual decides to act in a certain way, this action must be morally embraced by all rational beings when they encounter similar situations.

There are four different formulations of categorical imperative (see [31, 32] for a detailed discussion), but the main idea is that the pure practical rule is the main moral principle through which we can reflect on our actions from a moral point of view and assess their moral status. Let’s clarify this rule with an example: let’s say I am thinking of donating money to a charity. This is my subjective maxim, namely my personal idea for an action. Then, I am thinking about the moral status of this action and consider the categorical imperative. It means that I am asking this question: “can donating money to a charity be a universal law for every rational being?”. If my intention in formulating this maxim is “helping others” rather than, for instance, “to be known as a generous person in society”, then this action is morally permissible. Of course, there are several nuances that must be considered in the process of assessment (e.g., I have to do research into the charity to be sure that they are not a fraud organization, etc.), but we can briefly exemplify it in this fashion.

We can move to the next question: Why must one obey this universal rule? Kant finds the answer for this question again in the principles of rationality. Kant begins the Critique of Practical Reason by stating that “there is only one thing in the world which can be good without any limitation: goodwill” ([14], 4:393). Shortly, this good will is the vessel of rational moral law that reminds itself to the agent in every circumstance as the call of morality. If the will is determined by rules of the moral law, then it is a goodwill. The call of goodwill is a kind of constraint on desires and the main and only motivation for a real moral action. This limitation and call are the idea of duty ([3], p. 11). In a way, duty identifies the universality and necessity of the moral law. Following the moral law is not a choice of the absolute free will, but the only legitimate way to exist as a rational being. In other words, if one wants to exist as a human being, he/she has to follow the duty and do what is rationally right. For Kant, acting morally from self-interest or for seeking happiness is not linked to goodwill. Goodwill only appears when a subject acts from duty (doing what is rationally right). Any other motivation might conform with the duty, but the proper moral action comes from duty. There is only one legitimate motivation that is linked to goodwill and moral law: respect. We, as rational beings, feel respect for the authority of the pure moral law over all rational beings and also respect those who can carry this moral law in themselves ([8], p. 178).

At the end of all this, one can still say that “well, okay, I understand the moral law, the duty and also feel the respect for the moral law but I do not care.” Kant is crystal clear, as we repeat, that morality must come from rational principles and not based on prudential inclinations, because only in this way can we reach the highest good, which makes us worthy of happiness at the end ([14], 5:111). Otherwise, we would be like a robot that acts only based on the laws of nature and there would be no way to act through freedom ([17], p. 24). Acting as a free being requires us to act in accordance with moral duty because there is no other inclination or motivation

rather than goodwill, which is not prudential but rational. In this way, the moral quest is an unending process. That is to say, there is no way to say at a certain point that “this is the best way to live a moral life and there is no need for further consideration”. This point is indicated in another formulation of the categorical imperative: “act as if you were through your maxims a law-making member of a kingdom of ends” ([16], 4:434). Kant positions each rational being as the member of the kingdom of ends, which represents the harmonious togetherness of rationally and morally responsible beings. In this sense, when we consider our actions in terms of their moral value, we kind of envision an order in which every rational being follows this action and keeps the moral harmony at the same time. Even though it is a matter of discussion in the literature, the kingdom of end is an ideal to which we always aim to approach but never fully reach. This formulation expresses the constancy and the communal side of moral duty ([18], p. 192). Hence, Kant’s moral theory does not propose following moral rules in a secluded personal world but requires maintaining the moral public order where we can realize our actions morally. In summary, Kant’s moral theory postulates that humans must follow the supreme moral law identified through the categorical imperative as the source of the moral duty and the object of our respect in order to live in a harmonious moral world. In our conceptualization, we argue that Kant’s position on morality applies to both physical and virtual worlds, on which we elaborate in the next section.

4 A KANTIAN FRAMEWORK FOR THE METAVERSE

The confusion in the discussion of VR ethics directly links with the unsettled nature of VR [22]. That is to say that since it is not easy to decide the reality status of VR, it is not easy to develop ethical regulations in VR. However, in terms of shared social realms, it might not be necessary to attain the reality status of VR and then decide the ethical regulations. Even beyond that, **the dynamic of social realms in VR might not be much different from our actual reality.** Kant reserves the Critique of Pure Reason to limit our knowledge with the phenomenal world and explains the reasons why we cannot extend our knowledge to non-empirical concepts (such as the goodness in our example). If there is no way to theoretically know the noumenal world (the physical world), then there is no way to have the positive ontological determination of morally essential concepts and then organize the moral regulations. For this reason, Kant develops his moral understanding based on the working principles of human reason ([16], 4:404). In this approach, Kant asserts that **we do not realize the good, but the good emerges as a result of our rational action.**

Based on the foregoing discussion, it might seem that Kant’s philosophy means little to VR ethics. However, when we consider the nature of human beings and their knowledge from this point of view, the link becomes more conspicuous. In Kant’s paradigm, our knowledge begins with empirical experience ([13], A2). The knowledge of the phenomenal world, where the experience occurs, is subject to the operation of the faculties of the mind, e.g., sensibility, understanding, and imagination. This means that **we can know things only as they appear to us and there is no way to know how or what they are in themselves** ([32], p. 63). Therefore, we

do not have any access to the knowledge of objects beyond their appearances, because there is no way to extend the operation of the mind beyond the phenomenal world. In this regard, **the status of objects in VR does not differ from the real world because, in both worlds, objects are nothing other than the organized collection of the representations of the mind** ([30], p. 172). This also applies to our interactions with other human beings. When one encounters another human being, the operation of the mind for cognizing this being is identical to the process of knowing applied to objects. As two individuals are interacting with one another, their knowledge of the other is limited to their physical appearances, which, based on empirical knowledge, leads them to recognize one another as fellow human beings. On the other hand, in virtual environments, the perspective that legitimizes every sort of action over the avatars operates under the following assumption: “it is not the real world, and none of us here is a real person, so there is no harm caused to anyone because my actions does not really affect a real being”. That said, in the above-mentioned way of thinking, there is no way to know the avatar in itself; **one can only have the following judgment about the virtual worlds in terms of the reality of the human being, avatars, and the virtual objects:**

The Kantian judgment of reality in VR:

“I assume that I am a real person and I have an avatar in the VR world as an extension of my real being. For others, I only see things that look like my avatar, and my experience with these appearances leads me to the assumption that these avatars are likely extensions of human beings in the real world.”

The theoretical discussion so far gives us the ground for an ethical framework. At this point, we can summarize the theoretical inference in the following way: there is no way to give a clear-cut response to the reality question regarding the VR environments. That said, **(i) we know that there is a real intersubjective experience in these environments and that the reality of experience in virtual worlds does not differ from the reality of experience in the real world.** By applying Kant’s framework to this environment, there is no way to know the real nature of the encountered avatar theoretically, as, in the real world, the person in front of me is only an object appearing to me. On the other hand, **(ii) we know that our avatars are an extension of our real beings, and our experiences through my avatar in VR environments have effects on our real selves.** In this regard, our avatars, which are a part of us, are capable of having goodwill. Thus, goodwill, as its nature requires, imposes constraints on the individual to act morally, because it is determined by the supreme moral law, namely categorical imperative. If we carry the good-will and the necessary rational capacity to consider the categorical imperative, then it means that we, even as avatars, are subject to the duty (i.e., the responsibility to do what is rationally right.) **(iii) If this is the condition of our avatars, we must assume that other avatars must have the same capabilities regardless of being an extension of a real person, or even an AI-based character, because we can only know them as they appear to us, not in themselves.** And they appear to us as avatars like ourselves. It means that we can never treat them as means, but we must treat them as an end in themselves. Therefore, we must respect them because they seem

capable of carrying this moral law in themselves. **It, therefore, follows that we must act morally in virtual worlds as we do in real world.**

5 DISCUSSION

In this paper, we proposed an ethical framework (a method of approaching ethical issues) for social VR environments based on Kant’s Theory of Morality. The basic premise of this theoretical paper is that Kant’s theoretical and practical philosophy offers a robust foundation for approaching ethical issues in social VR. We argue that the reality of VR is not theoretically different from the reality of real life and that what is morally unacceptable in real life is and should be unacceptable in social VR. Thus, the **categorical imperative**, as the supreme moral law, is applicable to social VR, where users carry the **goodwill** through their avatars. As extensions of rational human beings, avatars are worthy of **respect** and are subject to the **duty**. These four concepts are deeply interwoven in forming the grounds of our Kantian framework for social VR. Therefore, violation of one them constitutes a violation of all others, rendering an action immoral according to Kant’s moral theory.

The emphasis of Kantian duty ethics on universal principles and respect for individual autonomy is particularly applicable to the unique ethical challenges posed by social VR. By prioritizing the dignity and rights of all users, Kant’s approach can contribute to ensuring that social VR environments are inclusive, respectful, and conducive to positive social interaction.

It is important to reiterate that Kant’s theory of morality is not the only ethical framework through which we can approach ethical issues in social VR. Consequentialism [5, 21] and/or virtue ethics [1, 2]—two other main theories in normative ethics—can also be applied to social VR. The merit of Kant’s morality is that Kant’s theoretical and practical philosophy provides simple and consistent answers for reality and ethics question for virtual worlds. Additionally, Kant’s theory of morality, also denoted as deontology, is grounded on rules rather than moral character, as in virtue ethics, or consequences of actions, as in consequentialism or utilitarianism. That’s why, it is more appropriate for and applicable to developing a common ethical understanding that could be generalized to the vast variety of issues in social VR.

5.1 Ethical Principles for Social VR

Based on our framework, we now present three main ethical principles derived from Kant’s theory of morality. These principles are formulas without any empirical content and provide a mechanism for decision making regarding ethical issues in social VR:

- (1) “Categorical Imperative” is valid for social VR and there is no way to legitimize the idea that virtual worlds are places where people can act in total freedom without considering any ethical rules.
- (2) One must assume that the encountered avatar is an extension of a being who has goodwill and worthy of respect.
- (3) One must act in accordance with duty in social VR environments

5.2 Practical Guideline Examples

In this section, we illustrate how these three principles can serve as the basis for developing practical ethical guidelines about issues pertinent to social VR. Following these three principles, we propose a set of practical guidelines just as examples. The list of guidelines is, by no means, complete or monolithic. On the contrary, these guidelines are open to change and they can be extended or modified for different circumstances and applications. The main purpose of presenting these exemplary guidelines is to show that it is possible to derive practical guidelines from these three principles without limiting ourselves with any determined set of rules.

- (1) Virtual harassment, groping, identity theft, rape, torture, fraudulence and any other action that is not ethically allowable in the real world cannot be allowed in virtual worlds.
- (2) Designers and developers should not generate social VR environments in which people can perform these illicit actions.
- (3) Users must be warned about the importance of being mindful of ethical concerns in virtual worlds.
- (4) Those who violate the rights of an individual should be held accountable for their actions both in virtual worlds and in real life, if necessary.
- (5) Designers and users must uphold the principles of morality when creating and using social VR applications.

It should be noted, however, that in the Kantian ethical framework, it is not possible to theoretically claim that a particular practical action is morally obligatory and must be obeyed under every circumstance. Even Kant himself gives examples based on concrete actions like suicide and lying but also highlights the point that defending any sort of statement like “do X” or “do not Y” as the ideal moral rule where every person must obey without even reasoning is not in accordance with his philosophy. Kant postulates that such a claim and way of acting would cause fanaticism because it means to claim that it is possible to determine the content of the moral law beyond its form and someone can have this ability ([14], 5:136). It would make the ethical reasoning futile and unnecessary. This might sound uncanny, but Kant himself does not propose any sort of universal statement with an empirical content. In this way, morality and moral reasoning is a subjective process that finds its objectivity in the shared capacity of moral rationality. That is why, it is not reasonable to make broad statements like “this application is ethically allowable, but that application is not”, as they would not be proper in an ethical reasoning following Kantian morality. Rather, Kant’s framework is appropriate for ethical reasoning at the individual level. In the next section, we illustrate how our framework could be applied when reasoning about ethical issues in social VR through three case studies.

5.3 Case Studies

Given the focus of our paper, we further discuss our contributions by illustrating how our framework based on Kantian ethics can be applied to decision making about issues relevant to social VR. For this purpose, we provide three case studies and explain how we conceptualize the application of these Kantian principles as a basis for ethical reasoning in social VR.

Case 1: Let’s assume that there is a user called Persona A (Figure 1a), who wants to join social VR environments. Persona A has

a maxim that can be formalized in the following way: “there are some actions that I want to try or perform but are also morally or legally prohibited in the real world. Since virtual worlds are not real, I can do them in this world.” As we discussed before, every agent naturally has a maxim, namely a subjective idea or expression of intentions, in engaging an action. In Kantian morality, an agent must examine the maxim in terms of its accordance with the categorical imperative to decide whether this action is morally permissible, obligatory, or prohibited. Let’s examine Persona A’s maxim in light of the developed framework.

Persona A’s maxim does not indicate a specific action. It might also have an addition like “so, I can harass someone verbally or physically”. What we need to look at is not the specific action but the principle that shaped the maxim. This maxim is clearly shaped by the principle of self-satisfaction or self-love (for Kant’s case example on self-love, see [16], 4:422). There is no consideration of the possible harm to others. In the second part of the maxim, the legitimization of the idea in the first part comes from the idea of virtuality: if none of this is real, then everything is permissible. However, as discussed above, there is no way to justify the causal relation between the virtuality of the world and the idea of total freedom. Also, there is no theoretical proof for knowing the reality status of the virtual world for sure. Therefore, this maxim cannot be considered as one that leads to an ethically permissible action.

Case 2: Let’s now consider the case of Persona B (Figure 1b). Persona B is someone who does not have good social relations in the real world. So, Persona B shapes a maxim in the following way: “I do not have good social relations in the real world. Virtual worlds afford the chance of establishing social relations independent from the real world. Therefore, I will spend all of my time in virtual worlds and will not spend any time in social life in the real world”. As you can see, this maxim is different from the one in the first example because in this example there is no violation of the rights of others. And, at first glance, it seems that there is a personal choice regarding spending spare time. When we consider the principle that shapes the maxim, nevertheless, we see that there is a moral commitment to the maxim.

As we discussed before, for Kant, happiness and benefit cannot be at the center of one’s principle of moral action because they are subjective practical principles through which there is no possible way to have objective criteria for morality. In this example, Persona B forms a maxim to compensate for his/her unhappiness in social life in the real world and decides to rather isolate from the real world to move his/her social experience to virtual worlds. Since the leading principle in this maxim is maximizing happiness and minimizing dissatisfaction, which are subjective and thus cannot be universal, this action is not ethically permissible based on Kant’s framework. On the other hand, if Persona B’s principle of maxim was based on a desire to maintain and/or improve their mental health because they suffer from severe social interaction anxiety in the real world, then the maxim would be universalizable for every rational being and therefore would in fact be permissible—and even obligatory in some instances—according to Kant’s theory of morality.

Case 3: Persona C (Figure 1c) is someone who spends considerable time in virtual worlds for social and professional engagements and has the following maxim: “If I were to use someone else’s name

and avatar, I could have better social and work relations and access some information that I could not have when I use my real identity. So, I will use someone else's identity or a fake identity when I feel it is more advantageous". In the case of using someone's identity, it might be clear to everyone that this action is not morally and legally permissible. However, in the case of using a fake identity, the decision seems more complicated. Nonetheless, if we look at the maxim in terms of its core principle, both cases will be obvious. The leading principle of the maxim is self-interest. Persona C sees actions maximizing self-interest as worthy of pursuing. Self-interest cannot be based on moral duty and respect because acting in this way is against the principle of universalizability (i.e., what is in Persona C's self-interest may not be generalized to everyone else) and the principle of seeing others as ends rather than means. Therefore, regardless of whether it is identity theft or using a fake identity, it is not ethically allowable in our framework.

6 CONCLUSION

In this paper, we proposed a Kantian ethical framework for social VR environments. By applying the principles of Kant's Theory of Morality, including the categorical imperative, goodwill, respect, and duty, we have demonstrated how ethical behavior in social VR can be evaluated and guided by a set of clear and consistent moral principles. Our Kantian framework provides a comprehensive and robust approach to assessing moral situations in social VR. As the field of HCI continues to evolve and expand, it is essential that we continue to develop theoretical frameworks that address the unique ethical challenges presented by emerging technologies. We hope that this paper will inspire further research and discussion on the ethical issues in social VR and contribute to the development of ethical guidelines that promote positive and responsible use of social VR environments.

REFERENCES

- [1] Julia Annas. 2006. Virtue ethics. In *The Oxford Handbook of Ethical Theory*, David Copp (Ed.). Oxford University Press, 515–536.
- [2] Aristotle. 2014. *Aristotle: Nicomachean Ethics*. Cambridge University Press.
- [3] Anne Margaret Baxley. 2010. *Kant's Theory of Virtue: The Value of Autocracy*. Cambridge University Press.
- [4] Katharina-Maria Behr, Andreas Nosper, Christoph Klimmt, and Tilo Hartmann. 2005. Some practical considerations of ethical issues in VR research. *Presence* 14, 6 (2005), 668–676.
- [5] Jeremy Bentham. 1961. *An Introduction to the Principles of Morals and Legislation*. Doubleday.
- [6] Philip Brey. 1999. The ethics of representation and action in virtual reality. *Ethics and Information Technology* 1, 1 (1999), 5–14.
- [7] David J Chalmers. 2022. *Reality+: Virtual worlds and the problems of philosophy*. Penguin UK.
- [8] Stephen Darwall. 2008. Kant on respect, dignity, and the duty of respect. In *Kant's Ethics of Virtue*, Monika Betzler (Ed.). de Gruyter, 175–200.
- [9] Julia Driver. 2005. Normative ethics. *The Oxford handbook of contemporary philosophy* (2005), 31–62.
- [10] Carlos Bermejo Fernandez and Pan Hui. 2022. Life, the Metaverse and Everything: An Overview of Privacy, Ethics, and Governance in Metaverse. *arXiv preprint arXiv:2204.01480* (2022).
- [11] Rosalind Hursthouse. 1999. *On virtue ethics*. OUP Oxford.
- [12] Robin Kabha. 2019. Ethical Challenges of Digital Immersive and VR. *Journal of Content, Community & Communication* 9 (2019), p41–49.
- [13] Immanuel Kant. 1998. *Critique of Pure Reason*. Cambridge University Press. <https://doi.org/10.1017/CBO9780511804649>
- [14] Immanuel Kant. 2015. *Critique of practical reason*. Cambridge University Press.
- [15] Immanuel Kant. 2017. *Kant: The metaphysics of morals*. Cambridge University Press.
- [16] Immanuel Kant. 2020. Groundwork of the Metaphysic of Morals. In *Immanuel Kant: Groundwork of the Metaphysic of Morals in focus*. Routledge, 17–98.
- [17] Markus Kohl. 2016. Kant on idealism, freedom, and standpoints. *Archiv für Geschichte der Philosophie* 98, 1 (2016), 21–54.
- [18] Christine M Korsgaard and Christine Marion Korsgaard. 1996. *Creating the kingdom of ends*. Cambridge University Press.
- [19] Michael Madary and Thomas K Metzinger. 2016. Real virtuality: A code of ethical conduct. Recommendations for good scientific practice and the consumers of VR-technology. *Frontiers in Robotics and AI* (2016), 3.
- [20] Joshua McVeigh-Schultz, Elena Márquez Segura, Nick Merrill, and Katherine Isbister. 2018. What's It Mean to "Be Social" in VR? Mapping the Social VR Design Ecology. In *Proceedings of the 2018 ACM Conference Companion Publication on Designing Interactive Systems*. 289–294.
- [21] John Stuart Mill. 1998. *Utilitarianism*. Oxford University Press.
- [22] Norman Mooradian. 2006. Virtual reality, ontology, and value. *Metaphilosophy* 37, 5 (2006), 673–690.
- [23] Stephen Rainey and Philippe Goujon. 2011. Toward a normative ethics for technology development. *Journal of Information, Communication and Ethics in Society* 9, 3 (2011), 157–179.
- [24] Katie Shilton et al. 2018. Values and ethics in human-computer interaction. *Foundations and Trends® in Human-Computer Interaction* 12, 2 (2018), 107–171.
- [25] Ketaki Shriram and Raz Schwartz. 2017. All are welcome: Using VR ethnography to explore harassment behavior in immersive social virtual reality. In *2017 IEEE Virtual Reality (VR)*. IEEE, 225–226.
- [26] Walter Sinnott-Armstrong and Edward N Zalta. 2015. Consequentialism. The stanford encyclopedia of philosophy.
- [27] Mel Slater, Cristina Gonzalez-Liencres, Patrick Haggard, Charlotte Vinkers, Rebecca Gregory-Clarke, Steve Jelley, Zillah Watson, Graham Breen, Raz Schwarz, William Steptoe, et al. 2020. The ethics of realism in virtual and augmented reality. *Frontiers in Virtual Reality* 1 (2020), 1.
- [28] Edward H Spence. 2008. Meta Ethics for the Metaverse: The Ethics of Virtual Worlds. *Current Issues in Computing and Philosophy* 175 (2008), 3.
- [29] James S Spiegel. 2018. The ethics of virtual reality technology: Social hazards and public policy recommendations. *Science and engineering ethics* 24, 5 (2018), 1537–1550.
- [30] Derek Stanovsky. 2004. Virtual Reality. In *The Blackwell Guide to the Philosophy of Computing and Information*, Luciano Floridi (Ed.). Blackwell, 167–177.
- [31] Allen W Wood. 1999. *Kant's ethical thought*. Cambridge University Press.
- [32] Allen W Wood. 2005. *Kant (Blackwell great minds)*.